2021-Summer Paper 1 (practice) – Basic Sciences in Urology

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Practice paper compiled from the question bank; the actual 2021-Summer Paper 1 was not available.

Q1. Anatomy of lymphatic drainage of testis and scrotum. Describe management implications of penile malignancies.

Type: LAQ (25 marks) (also answered in 2018-Summer Paper 2, Q2 [testis and scrotum] and 2017-Summer Paper 2, Q1 [inguinal nodes in penile cancer])

Second part taken as the management implications of lymphatic drainage in penile cancer, i.e. management of the inguinal and pelvic nodes.

Embryological basis

- The testis develops at L1–L2 and descends with its vessels and lymphatics, so its lymph returns to the **retroperitoneum**. The scrotum is skin and drains to the **groin**.

Lymphatic drainage of the testis

- Lymphatics of the testis, caput and corpus epididymis run up the **spermatic cord** with the gonadal vessels to the **para-aortic and interaortocaval nodes** near the renal hilum; then cisterna chyli, thoracic duct, **left supraclavicular** nodes.
- **Cauda epididymis** drains with the vas to the **external iliac** nodes.

	Right testis	Left testis
Primary landing zone	**Interaortocaval**	**Left para-aortic**
Secondary	Paracaval, precaval	Pre-aortic, interaortocaval
Crossover	Right to left occurs	Rare to right (bulky disease only)

- Crossover is right to left because lymph flows that way.

Lymphatic drainage of the scrotum

- Scrotal lymphatics **do not cross the median raphe**; they drain to the **ipsilateral superficial inguinal nodes**.

Lymphatic drainage of the penis

Superficial system	Deep system
Prepuce and shaft skin	Glans, frenulum, corpora
Dorsal network divides at the base → both superficial inguinal groups	Trunks circle the corona, run under Buck's fascia → presymphyseal plexus → femoral triangle nodes

- **Two rules:** drainage is **bilateral and unpredictable in side** (presymphyseal crossover), but **orderly in sequence**: superficial inguinal → deep inguinal → external iliac and obturator. **No skip metastases.**
- **Superficial nodes** (4–25, above fascia lata) lie in **Daseler's five zones**; sentinel nodes are mostly **superomedial**, then central. **Deep nodes** (1–3) lie medial to the femoral vein; **Cloquet's node** is the highest.

Clinical implications: testis and scrotum

- **Never operate trans-scrotally** on a testicular tumour. Scrotal violation opens skin lymphatics, so tumour can reach inguinal and pelvic nodes; use **radical inguinal orchidectomy** with high cord clamping.
- Inguinal or pelvic spread from the testis means altered drainage: scrotal violation, previous orchidopexy or hernia repair, scrotal invasion.
- RPLND (retroperitoneal lymph node dissection) templates and para-aortic or "dog-leg" radiotherapy fields follow the landing zones.

- Scrotal skin cancer is staged at the ipsilateral groin.

Management implications in penile cancer

1. Prognosis and timing

- Nodal status is the **strongest prognostic factor**. Penile SCC (squamous cell carcinoma) has a long **locoregional phase**, so the groin is a **curable station**.
- **Kroon (2005):** early ILND (inguinal lymph node dissection) for impalpable nodes gave 3-year disease-specific survival about **84% vs 35%** when delayed until palpable.
- Manage both groins **within 3 months**; remove **at least 8 nodes per side**.

2. Bilateral management

- Crossover means **both groins** are staged or treated; with one palpable groin, the other holds micrometastasis in about 30%.

3. Impalpable groin (cN0): risk-based

Risk	EAU-ASCO (European Association of Urology–American Society of Clinical Oncology) 2023	Groin
Low	pTis, pTa, pT1a G1	Surveillance
Intermediate	pT1a G2	Surveillance if compliant, or surgical staging
High	**pT1b or higher**, G3	**DSNB (dynamic sentinel node biopsy)**; ILND if DSNB unavailable

- NCCN (National Comprehensive Cancer Network): pT1b G1–2 is intermediate risk and is staged surgically (ILND first).
- **DSNB:** ultrasound ± FNAC (fine-needle aspiration cytology) first, because a tumour-replaced node diverts flow (false negative). Then technetium-99m nanocolloid, blue dye and gamma probe.
- **No-skip spread and superomedial sentinels** justify **modified ILND (Catalona)**: smaller field, saphenous vein preserved; positive frozen section → complete radical ILND.

4. Palpable nodes (cN+)

- About half are reactive, but do not delay with antibiotics. Ultrasound-guided **FNAC**; positive → **radical ILND** on that side, stage the other.
- **Bulky (≥4 cm), fixed or bilateral nodes:** neoadjuvant **TIP** (paclitaxel, ifosfamide, cisplatin) × 4, then surgery in responders.

5. Pelvic nodes

- Stepwise spread means a negative groin needs no pelvic dissection.
- **PLND (pelvic lymph node dissection)**, ipsilateral, if **more than 2 inguinal nodes or ENE (extranodal extension)**. ENE in any node is pN3.

6. Complications of ILND

- Wound infection, **flap necrosis**, **lymphocele**, **lymphoedema**, DVT (deep vein thrombosis). Reduced by modified templates, saphenous sparing and **VEIL (video-endoscopic inguinal lymphadenectomy)**.

Also know: adjuvant therapy for pN2–3; Cabanas sentinel biopsy abandoned.

Diagram to draw: retroperitoneal landing zones (right interaortocaval, left para-aortic) beside the femoral triangle with Daseler's zones and Cloquet's node.

> **Key points to remember:**
>
> - Right testis → interaortocaval; left → para-aortic; crossover right to left only.
> - Scrotum and penis → inguinal nodes; never operate trans-scrotally.
> - Penile drainage is bilateral and stepwise without skips: stage both groins, PLND only if inguinal nodes positive.
> - pT1b or higher: DSNB or ILND within 3 months; Kroon 84% vs 35%.

Q2. Describe IVU & its applications in Urology.

Type: LAQ (25 marks)

Definition and principle

- IVU (intravenous urography; also IVP, intravenous pyelography or excretory urography) is a series of timed abdominal radiographs after intravenous iodinated contrast.
- Contrast is **freely filtered by the glomerulus and not reabsorbed**. As water is reabsorbed it is concentrated in the tubules (**nephrogram**) and then outlines the calyces, pelvis, ureters and bladder (**pyelogram**).
- So IVU shows **anatomy plus a rough, qualitative idea of function and drainage** of each kidney.
- **Current status:** largely replaced by CTU (CT [computed tomography] urography). It is still used where CT is unavailable (common in India) or when dynamic excretion and drainage must be seen.

Contrast media

Type	Example	Feature
High-osmolar ionic	Diatrizoate	More reactions; obsolete for IV use
Low-osmolar non-ionic (standard)	Iohexol, iopamidol	Fewer reactions
Iso-osmolar non-ionic	Iodixanol	High-risk renal patients

Preparation

- Check creatinine or eGFR (estimated glomerular filtration rate), allergy, asthma, pregnancy, diabetes on metformin, myeloma.
- Laxative the night before (bowel gas hides the kidneys); short fast but **no dehydration**.
- Premedicate high-risk patients, e.g. prednisone 50 mg at 13, 7 and 1 h plus diphenhydramine 50 mg at 1 h.

Technique: film sequence

Film	Timing	Shows
Scout (control) KUB (kidney–ureter–bladder)	Before contrast	Calcification, bones, gas, technique
Contrast	Low-osmolar, about 1 mL/kg (usually 50–100 mL) IV	—
Nephrogram / nephrotomogram	About 1 min (60–90 s)	Renal size, outline, parenchyma
Pyelogram	5 min	Calyces; then apply **ureteric compression**
Compression film	10 min	Well-distended calyces and pelvis
Release (full-length) film	About 15 min	Whole ureters
Bladder and **post-void** films	20–30 min	Bladder outline, residue, lower ureters
Delayed films	Up to 24 h	Level of obstruction; **no excretion at 24 h = non-functioning kidney**

- **Also know:** films in expiration; prone film fills the ureters; oblique views; erect film for nephroptosis.

- Compression is avoided in suspected obstruction or colic, aortic aneurysm, recent abdominal surgery and trauma.

Contraindications

- Previous severe contrast reaction; significant renal impairment; pregnancy (relative); dehydration, myeloma.

Nephrogram patterns

Pattern	Causes
Absent	Renal artery occlusion, agenesis, avulsion
Delayed, increasingly dense	**Acute obstruction**, renal artery stenosis
Striated	Acute pyelonephritis, obstruction
Rim (cortical rim) sign	Renal infarction (capsular collaterals)
Bilateral persistent dense	Hypotension, ATN (acute tubular necrosis)

Applications in urology

Condition	IVU features
Calculi	Confirms an opacity is in the tract; **radiolucent stone = filling defect**; site, obstruction, calyceal anatomy before SWL (shock wave lithotripsy) or PCNL (percutaneous nephrolithotomy)
Acute obstruction	Delayed dense nephrogram, delayed excretion, dilatation down to the block, **standing column** of contrast, forniceal rupture
PUJ (pelviureteric junction) obstruction	Ballooned pelvis, delayed drainage, non-dilated ureter; crossing-vessel indentation
Duplex system	Upper moiety obstructed or non-functioning: lower moiety displaced, "**drooping lily**"
Ureterocele	"**Cobra head**" (spring onion) at the bladder
Horseshoe kidney	Low, medially rotated; lower calyces point medially
Retrocaval ureter	"Reverse J" or **fish-hook** at L3
Medullary sponge kidney	"**Paintbrush**" papillary striations
Papillary necrosis	**Ring sign**, "lobster claw" calyces
Genitourinary TB (tuberculosis)	**Moth-eaten** calyx, **phantom calyx**, Kerr's kink, **pipestem** or beaded ureter, contracted bladder, autonephrectomy
Urothelial tumour	Irregular filling defect, non-visualised calyx
Retroperitoneal fibrosis	Medial deviation and tapering of ureters with hydronephrosis
Pelvic lipomatosis	Pear-shaped bladder
Schistosomiasis	Ureteric striations, calcified bladder

- **Trauma: one-shot intraoperative IVP:** 2 mL/kg IV push, single film at 10 min. It confirms a functioning contralateral kidney before exploration in the unstable patient.
- **Also know:** nephroptosis (descent >5 cm or two vertebral bodies supine to erect); urinary fistulae (ureterovaginal); renal masses (distortion); post-operative assessment; diverticula.

Complications

- **Contrast reactions:** immediate (urticaria to anaphylaxis) or delayed (rash). Severe reaction: stop contrast, **IM (intramuscular) adrenaline 0.3–0.5 mg** every 5–15 min

(maximum 1 mg), oxygen, β2-agonist. Observe 30 min.
- **CA-AKI (contrast-associated acute kidney injury):** prevent with hydration, low or iso-osmolar contrast, lowest dose.
- Radiation; metformin-related lactic acidosis in renal impairment.

Limitations and current role

- Two-dimensional; bowel gas; poor parenchymal detail; misses small masses; poor in renal failure.
- CTU is more sensitive for stones, masses and urothelial lesions, but gives a higher radiation dose than IVU. IVU remains useful in resource-limited settings.

Recent advances

- Split-bolus CTU (fewer phases); dual-energy CT (virtual unenhanced images); MR urography without radiation.

Diagram to draw: normal IVU with minor and major calyces, pelvis, ureter and the three narrowings; beside it a "drooping lily".

> **Key points to remember:**
> - Contrast is filtered, not reabsorbed: nephrogram then pyelogram.
> - Delayed, increasingly dense nephrogram = acute obstruction; no excretion at 24 h = non-functioning.
> - Classic signs: drooping lily, cobra head, fish-hook, paintbrush, ring sign, phantom calyx.
> - Severe reaction: IM adrenaline 0.3–0.5 mg; CTU has replaced IVU.

Q3a. Premalignant conditions in urology

Type: SAQ (10 marks)

Definition

- Altered tissue with a raised risk of invasive cancer, often **intraepithelial neoplasia with an intact basement membrane**. Treating it prevents cancer.

Penis

Lesion	Key points
PeIN (penile intraepithelial neoplasia)	HPV (human papillomavirus)-associated (commonest) or differentiated (HPV-independent). WHO (World Health Organization) 2022: all PeIN is high grade
Erythroplasia of Queyrat	Red velvety glans plaque; **progresses in 10–33%**
Bowen disease	Scaly shaft plaque; about 5% progress
Bowenoid papulosis	Shaft papules in young men; does not progress
Lichen sclerosus (BXO, balanitis xerotica obliterans)	SCC (squamous cell carcinoma) in **2.3–9%**, HPV-independent type
Giant condyloma (Buschke–Löwenstein)	Locally invasive; may harbour SCC
Also know	Cutaneous horn, micaceous balanitis, phimosis, genital lichen planus

- **Mechanism (HPV):** E6 inactivates p53, E7 inactivates retinoblastoma protein; p16 is the marker.
- **Treatment:** biopsy first; circumcision for preputial lesions; topical **5-fluorouracil or 5% imiquimod**; laser; **glans resurfacing**. Lichen sclerosus: clobetasol 0.05%, lifelong follow-up.

Bladder

- **CIS** (carcinoma in situ): flat high-grade lesion, umbrella cells lost; untreated, **40–83%** invade muscle. Treat with BCG (bacillus Calmette–Guérin).
- **Urothelial dysplasia:** 14–19% progress; 60% to CIS with prior cancer.
- **Cystitis glandularis** (intestinal type): low risk of adenocarcinoma; resect and survey.
- **Leukoplakia** (keratinising squamous metaplasia): traditionally premalignant; Campbell now says **not** (no TP53 changes). Schistosomiasis and chronic catheters predispose to SCC.

Prostate

- **HGPIN** (high-grade prostatic intraepithelial neoplasia): precursor; isolated HGPIN no longer needs repeat biopsy. IDC-P (intraductal carcinoma of the prostate) usually means invasive cancer nearby.

Testis

- **GCNIS** (germ cell neoplasia in situ): precursor of all postpubertal GCTs (germ cell tumours) except spermatocytic tumour; **50% invasive within 5 years**. Treat by orchidectomy or **radiotherapy ≥20 Gy**.
- **Gonadoblastoma** in dysgenetic gonads with Y material; cryptorchidism.

Kidney and others

- **Also know:** papillary adenoma; acquired cystic kidney disease on dialysis; urethral diverticulum; bowel in augmentation.

Diagram to draw: normal urothelium vs dysplasia vs CIS (umbrella cells lost).

Key points to remember:

- PeIN is all high grade; Queyrat 10–33% vs Bowen 5% progression.
- Untreated bladder CIS: 40–83% invade.
- GCNIS: 50% invasive in 5 years; radiotherapy ≥20 Gy.

Q3b. Bacteriuria in pregnancy.

Type: SAQ (10 marks) (also answered in 2019-Summer Paper 3, Q3b)

Definition

- **ASB (asymptomatic bacteriuria):** ≥10^5 CFU (colony-forming units)/mL of one uropathogen in a midstream sample without urinary symptoms (≥10^2 CFU/mL on a catheter sample).

Epidemiology

- Prevalence is **not raised** by pregnancy (about 2–10%); higher with low socioeconomic status, multiparity, sickle cell trait. Organisms as in non-pregnant women (mainly *E. coli*).

Why it matters (pathophysiology)

- **Progesterone** relaxes ureteric smooth muscle; the **uterus compresses the ureters at the pelvic brim** (right more). Dilatation from about 15 weeks; normal by 6 weeks postpartum.
- Result: **stasis and hydroureteronephrosis**; bacteriuria persists and ascends.
- GFR (glomerular filtration rate) rises 30–50%: creatinine and drug levels fall.
- **Untreated ASB → pyelonephritis in 15–45%** (historical), 60–75% in the third trimester. Treatment cuts this to **1–4%**.
- Pyelonephritis causes **preterm birth, low birth weight** and maternal sepsis. **Cochrane 2019:** treating ASB reduces all three.

Screening

- **Urine culture at 12–16 weeks** or the first visit (USPSTF [US Preventive Services Task Force], IDSA [Infectious Diseases Society of America]). Dipstick misses many cases.

Treatment

Safe	Restricted or avoided
β-lactams (amoxicillin, cephalexin): throughout	**Nitrofurantoin:** stop from 35 weeks (neonatal haemolysis, G6PD [glucose-6-phosphate dehydrogenase] deficiency); not for pyelonephritis
Fosfomycin: lower tract only	**Trimethoprim-sulfamethoxazole:** avoid 1st trimester (folate, neural tube); caution near term (kernicterus)
Gentamicin: pyelonephritis	**Fluoroquinolones**, **tetracyclines**, chloramphenicol: avoid

- ASB and cystitis: culture-directed, **4–7 days** (cures more than a single dose); **repeat culture** after treatment.

Pyelonephritis

- **Admit**; IV (intravenous) β-lactam (± gentamicin) until afebrile, then oral to complete ≥14 days. No improvement by **72 h → ultrasound** (stone, obstruction, abscess).

Prophylaxis

- After one pyelonephritis or two cystitis episodes: nightly nitrofurantoin or cephalexin until delivery, or close culture surveillance (Cochrane 2015).
- **Also know:** urinary diversion or augmentation, SCI (spinal cord injury): most treat ASB in pregnancy.

Diagram to draw: right hydroureter compressed at the pelvic brim.

Key points to remember:

- Prevalence unchanged; risk of pyelonephritis raised (15–45% untreated).
- Screen by culture at 12–16 weeks; treat 4–7 days.
- β-lactams safe; no fluoroquinolones or tetracyclines; nitrofurantoin not after 35 weeks.

Q3c. Complications if DJS

Type: SAQ (10 marks)

Taken as complications of a DJ (double-J) ureteric stent.

Introduction

- The **double-J (double-pigtail)** stent (Finney 1978) is a self-retaining polymer tube with coils in the pelvis and bladder. Consent should cover its symptoms and complications.

Classification of complications

Timing	Complications
At insertion	**Misplacement**: proximal end below the obstruction; **submucosal** passage or perforation (suspect when force is needed)
Early	**Stent-related symptoms**; haematuria; infection and urosepsis; **vesicoureteric reflux** up the stent (flank pain on voiding)
Late	**Migration** (proximal or distal; short stent); **encrustation**; **stent failure** (obstruction); biofilm and colonisation; **fracture**; **forgotten stent**; rarely **ureteroarterial fistula**

Stent-related symptoms

- About **4 in 5 patients**: urgency, frequency, dysuria, haematuria, flank and groin pain; sexual and work problems.
- **Mechanism:** the **intravesical coil irritates the trigone**, and **reflux up the stent** distends the kidney on voiding.
- Measured by **USSQ** (Ureteral Stent Symptom Questionnaire).
- **Reduced by: α-blockers and antimuscarinics**; intraureteral placement; hydrocoated silicone; shortest dwell. Diameter does not matter.

Encrustation and forgotten stent

- Minerals and biofilm deposit on the stent. Risk rises with **dwell time**, infection, stone-forming urine and pregnancy. Leads to obstruction, sepsis, fracture, **ureteric avulsion on removal**, renal loss.
- **FECal grading (Forgotten, Encrusted, Calcified):** I minimal linear; II circular; III partial on one or both coils; IV complete on one coil; V both coils.
- **Management:** non-contrast CT (computed tomography); never pull by force; clear **bladder coil** (cystolithotripsy), then **ureter** (ureteroscopy), then **renal coil** (flexible ureteroscopy or PCNL [percutaneous nephrolithotomy]).

Stent failure

- Extrinsic compression (malignancy, radiation), debris, encrustation. Obstructive symptoms → image; change stent, **PCN** (percutaneous nephrostomy) or **metallic stent**.

Prevention

- Stent only when indicated; correct length; **stent register**; long-term stents changed every 3–6 months (local practice); **pregnancy: about every 4 weeks**.

Diagram to draw: DJ stent in situ with bladder coil on the trigone and arrows showing reflux.

Key points to remember:

- Symptoms in about 80%: trigonal irritation and reflux; α-blockers and antimuscarinics help.
- Encrustation rises with dwell time; FECal I–V.
- Encrusted stent: bladder, then ureter, then kidney; keep a stent register.

Q3d. Proteinuria

Type: SAQ (10 marks) (also answered in 2017-Summer Paper 1, Q3f)

Definition

- Normal **80–150 mg/day** (albumin <30 mg/day). The glomerulus holds back albumin; the proximal tubule reabsorbs small proteins.
- **Nephrotic range:** >3.5 g/day with hypoalbuminaemia, oedema and hyperlipidaemia.

Types

Type	Mechanism	Amount	Example
Glomerular (commonest)	Leaky filter	Often >2 g; albumin about 70%	Diabetes, glomerulonephritis, amyloid
Tubular	Failed reabsorption of small proteins	Seldom >2–3 g; globulins	Fanconi syndrome, heavy metals, drugs
Overflow	Excess plasma protein	0.3–2 g	**Myeloma** (Bence Jones), myoglobin, haemoglobin

By timing	Features
Transient	Fever, exercise, heart failure; commonest in children
Orthostatic	Young men, upright only, <1 g; benign (normal first-morning sample)
Persistent	Mostly glomerular; needs full work-up

Detection and quantification

- **Dipstick** (tetrabromophenol blue) detects mainly albumin from 20–30 mg/dL; misses dilute urine and **light chains**. **Sulfosalicylic acid** precipitates all proteins; dipstick negative but sulfosalicylic acid positive → suspect myeloma.
- KDIGO (Kidney Disease: Improving Global Outcomes) categories by ACR (albumin-to-creatinine ratio): **A1 <30, A2 30–300, A3 >300 mg/g**. Alternatives: PCR (protein-to-creatinine ratio), 24-h urine.

Evaluation

```
Positive dipstick → repeat; exclude UTI, fever, exercise
→ persistent → quantify (ACR, PCR or 24-h)
→ creatinine/eGFR, sediment (dysmorphic RBCs, casts), ultrasound
→ glomerular (>2 g, albumin) → nephrology ± biopsy; C3, C4, ANA if systemic
→ globulins → electrophoresis, immunofixation (myeloma)
```

- UTI: urinary tract infection; eGFR: estimated glomerular filtration rate; RBC: red blood cell; ANA: antinuclear antibody.

Urological relevance

- **Haematuria work-up:** proteinuria with casts or dysmorphic cells suggests medical renal disease; urological bleeding adds little protein.
- **Reflux nephropathy:** proteinuria marks progression to renal failure.
- **After nephrectomy:** hyperfiltration causes proteinuria and FSGS (focal segmental glomerulosclerosis); favour nephron sparing.
- **Living donors:** significant proteinuria excludes donation. **Also know:** mTOR (mammalian target of rapamycin) inhibitors cause it.

Management

- Treat the cause; control blood pressure; **ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin receptor blocker)**; SGLT2 (sodium–glucose cotransporter-2) inhibitors.

Diagram to draw: glomerulus and proximal tubule marking glomerular, tubular and overflow sites.

> **Key points to remember:**
> - Normal <150 mg/day; nephrotic >3.5 g/day.
> - Glomerular, tubular, overflow; orthostatic is benign.
> - Dipstick misses light chains; use sulfosalicylic acid or immunofixation.

Q3e. Radionuclide studies for PUJ obstruction.

Type: SAQ (10 marks)

Introduction

- Dilatation is not obstruction. PUJ (pelviureteric junction) obstruction is impaired drainage that damages the kidney. **Diuretic renography** is the functional standard and best follow-up test.

Tracers

Tracer	Handling	Use
^{99m}Tc-**MAG3** (technetium-99m mercaptoacetyltriglycine)	Tubular secretion	**Agent of choice**: split function and drainage; good images even with poor function
^{99m}Tc-**DTPA** (diethylenetriaminepentaacetic acid)	Glomerular filtration	GFR (glomerular filtration rate); poor in renal impairment
^{99m}Tc-**DMSA** (dimercaptosuccinic acid)	Cortical retention	Static split function; no drainage data

"Well-tempered" diuretic renogram (standard protocol)

- **Hydration** (oral or IV [intravenous]): dehydration mimics obstruction.
- **Bladder catheter:** removes full-bladder and reflux effects.
- **Furosemide:** 0.5 mg/kg (maximum 40 mg) in adults; 1 mg/kg in children.
- **Timing: F+20** (20 min after tracer, when the pelvis is full), **F0** (with tracer) or **F−15** (15 min before, so the system is already in diuresis).
- Regions of interest over cortex and pelvis; then upright images.

Outputs

- **DRF (differential renal function)** from counts at 1–3 min; normal 45–55%.
- **Drainage half-time (T½)** from maximal filling:

T½ after furosemide	Interpretation
<10 min	Not obstructed
10–20 min	Equivocal
>20 min	Obstructed

- **O'Reilly curves:** I normal; II obstructed (keeps rising); IIIa dilated, non-obstructed (falls after furosemide); IIIb equivocal.
- **Also know:** P40 (activity remaining at 40 min); output efficiency; NORA (normalised residual activity).

Indications for pyeloplasty from renography

- **DRF <40%**, **fall in DRF >10%** on serial scans, **T½ >20 min**, upright retention; also symptoms, infection, increasing dilatation.

Pitfalls

- Dehydration, early diuretic, full bladder, wrong ROI (region of interest), very poor function.
- One Campbell chapter calls T½ >40 min obstructed; children have no agreed cut-off, so serial trends and curve shape matter.

Follow-up and alternatives

- After pyeloplasty: renogram 1 month after stent removal; spontaneous drainage means success.
- Equivocal → **Whitaker test** (>15–20 cm H_2O obstructed) or magnetic resonance urography.

Diagram to draw: normal, obstructed and dilated non-obstructed renogram curves.

> **Key points to remember:**
> - MAG3 is the tracer of choice; hydrate and catheterise.
> - DRF normal 45–55%; T½ <10 normal, >20 min obstructed.
> - Operate for DRF <40%, fall >10%, or poor drainage with symptoms.

Q3f. Immunosuppressants.

Type: SAQ (10 marks) (also answered in 2020-Summer Paper 4, Q1)

Taken as immunosuppressant drugs in renal transplantation.

Principle

- T-cell activation needs three signals: **signal 1** (TCR [T-cell receptor] binding donor antigen → calcineurin → IL-2 [interleukin-2] transcription), **signal 2** (costimulation, CD28–CD80/86) and **signal 3** (IL-2 receptor → mTOR [mammalian target of rapamycin] → proliferation). Each drug blocks a step.
- Phases: **induction** (peri-transplant), **maintenance** (lifelong), **rescue** (rejection).

Classification

Class	Drug	Mechanism	Main toxicity
CNI (calcineurin inhibitor)	**Tacrolimus**, ciclosporin	Block calcineurin (signal 1)	**Nephrotoxicity**, diabetes, tremor, hypertension; ciclosporin: gum hypertrophy, hirsutism
Antimetabolites	**Mycophenolate**, azathioprine	Block purine synthesis (IMPDH [inosine monophosphate dehydrogenase])	Diarrhoea, leucopenia
mTOR inhibitors	Sirolimus, everolimus	Block signal 3	**Lymphocele, poor wound healing**, proteinuria, dyslipidaemia; not nephrotoxic, antitumour
Corticosteroids	Prednisolone, methylprednisolone	Block cytokine transcription	Diabetes, bone loss, cataract
Costimulation blocker	**Belatacept**	CTLA4-Ig binds CD80/86 (signal 2)	**PTLD** (post-transplant lymphoproliferative disorder); avoid if EBV (Epstein–Barr virus)-negative
Depleting antibodies	rATG (rabbit antithymocyte globulin), alemtuzumab	T-cell depletion	Infection, PTLD
Non-depleting	**Basiliximab**	Anti-CD25 (IL-2 receptor)	Few
B-cell, antibody	Rituximab, IVIG (intravenous immunoglobulin), bortezomib, eculizumab, imlifidase	Antibody-mediated rejection, desensitisation	—

Standard regimen

- **Over 90%** receive induction and **tacrolimus plus mycophenolate**; steroids in more than two-thirds. Acute rejection has fallen from ≥60% to <10%.
- **Tacrolimus trough monitoring**; CYP3A4 (cytochrome P450 3A4) interactions (azoles, macrolides raise levels; rifampicin lowers).
- **Rejection:** T-cell-mediated → IV (intravenous) steroids, then rATG; antibody-mediated → plasmapheresis, IVIG, rituximab.

Urological relevance

- **Infection:** UTI (urinary tract infection), **BK virus** (ureteric stricture, nephropathy), CMV (cytomegalovirus).
- **Malignancy:** skin cancer, PTLD, RCC (renal cell carcinoma, mostly native kidneys), **bladder cancer about 3-fold** and more aggressive.
- **Surgery:** lymphocele and wound problems (mTOR); stones (CNI, steroids).
- **Also know:** belatacept (BENEFIT trial: better glomerular filtration rate), extended-release tacrolimus, TRANSFORM (everolimus with reduced CNI non-inferior).

Diagram to draw: antigen-presenting cell and T cell with signals 1–3 and the drug acting at each.

> **Key points to remember:**
> - Tacrolimus + mycophenolate ± steroids after induction (basiliximab or ATG).
> - CNI nephrotoxic; mTOR inhibitors cause lymphocele and poor healing.
> - Belatacept blocks signal 2; avoid if EBV-negative.

Rapid Revision Sheet – 2021-Summer Paper 1 (practice)

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Anatomy of lymphatic drainage of testis and scrotum. Describe management implications of penile malignancies. (LAQ)

- Right testis → interaortocaval; left → para-aortic; crossover right to left only.
- Scrotum and penis → inguinal nodes; never operate trans-scrotally.
- Penile drainage is bilateral and stepwise without skips: stage both groins, PLND only if inguinal nodes positive.
- pT1b or higher: DSNB or ILND within 3 months; Kroon 84% vs 35%.

Q2. Describe IVU & its applications in Urology. (LAQ)

- Contrast is filtered, not reabsorbed: nephrogram then pyelogram.
- Delayed, increasingly dense nephrogram = acute obstruction; no excretion at 24 h = non-functioning.
- Classic signs: drooping lily, cobra head, fish-hook, paintbrush, ring sign, phantom calyx.
- Severe reaction: IM adrenaline 0.3–0.5 mg; CTU has replaced IVU.

Q3a. Premalignant conditions in urology (SAQ)

- PeIN is all high grade; Queyrat 10–33% vs Bowen 5% progression.
- Untreated bladder CIS: 40–83% invade.
- GCNIS: 50% invasive in 5 years; radiotherapy ≥20 Gy.

Q3b. Bacteriuria in pregnancy. (SAQ)

- Prevalence unchanged; risk of pyelonephritis raised (15–45% untreated).
- Screen by culture at 12–16 weeks; treat 4–7 days.
- β-lactams safe; no fluoroquinolones or tetracyclines; nitrofurantoin not after 35 weeks.

Q3c. Complications if DJS (SAQ)

- Symptoms in about 80%: trigonal irritation and reflux; α-blockers and antimuscarinics help.
- Encrustation rises with dwell time; FECal I–V.
- Encrusted stent: bladder, then ureter, then kidney; keep a stent register.

Q3d. Proteinuria (SAQ)

- Normal <150 mg/day; nephrotic >3.5 g/day.
- Glomerular, tubular, overflow; orthostatic is benign.
- Dipstick misses light chains; use sulfosalicylic acid or immunofixation.

Q3e. Radionuclide studies for PUJ obstruction. (SAQ)

- MAG3 is the tracer of choice; hydrate and catheterise.
- DRF normal 45–55%; T½ <10 normal, >20 min obstructed.
- Operate for DRF <40%, fall >10%, or poor drainage with symptoms.

Q3f. Immunosuppressants. (SAQ)

- Tacrolimus + mycophenolate ± steroids after induction (basiliximab or ATG).
- CNI nephrotoxic; mTOR inhibitors cause lymphocele and poor healing.
- Belatacept blocks signal 2; avoid if EBV-negative.